

Readme Contents

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1. What is this?

The Bcalc Recoil Calculator is an open-source firearm recoil calculator.

You can enter specific firearm and ammo data and the calculator will estimate:

- free recoil energy: total energy absorbed
- recoil velocity: speed the firearm moves backwards
- average impulse force: average force on hands/ shoulder
- perceived recoil score: to be used in recoil comparisons

An internet connection is not required once installed. No user data is transmitted.

2. Who is this for?

Anyone interested in comparing recoil related calculations of different firearm and ammunition combinations.

3. Installation Guide

1. Linux

- download "bcalcrecoil" from the <https://github.com/briancalc/bcalcrecoil/releases> page
- to run using file manager, double-click bcalcrecoil
- to run from the terminal `./bcalcrecoil`

2. Windows

- in progress

3. Mac

- in progress

4. User Guide

Step 1. Enter a unique name for the data set if you want to be able to refer to it later without having to enter all the variables again. Not required however.

Input Parameters

Unique Name: 308 WIN

Cartridge: 308 Win

Firearm Class: rifle: standard

Action Type: bolt action

Gun Weight: 8.0 lbs

Bullet Weight: 125 grains

Muzzle Velocity: 2973 fps

Powder Charge: 46.5 grains

Muzzle Device: none

Stock Fit

LOP: just right

Comb Height: perfect cheek weld

Recoil Pad: soft recoil pad

Calculate **Save Data Set**

Results

	Data Set 1	Data Set 2	Data Set 3
Unique Name:	308 WIN	--	--
Cartridge Selected:	308 Win	--	--
Firearm Class:	rifle: standard	--	--
Action Type:	bolt action	--	--
Gun Weight:	8.0 lbs	--	--
Bullet Weight:	125.0 gr	--	--
Muzzle Velocity:	2973 fps	--	--
Powder Charge:	46.5 gr	--	--
Muzzle Device:	none	--	--
LOP:	just right	--	--
Comb Height:	perfect cheek weld	--	--
Recoil Pad:	soft recoil pad	--	--

Calculated Results

Free Recoil Energy	14.92 ft-lbs	-- ft-lbs	-- ft-lbs
Recoil Velocity	10.96 fps	-- fps	-- fps
Avg Impulse Force	1434 lbf	-- lbf	-- lbf
Perceived Recoil Score	5.7 /10	-- /10	-- /10





Step 2. Select cartridge, firearm class, and action type from the drop-down menus.

Step 3. Enter gun weight (lbs), bullet weight (grains), and muzzle velocity.

Step 4. Enter a powder charge weight (grains) if known. The calculator will estimate if left blank.

Step 5. Select muzzle device, LOP, comb height, and recoil pad from the drop-down menus.

Step 6. Press the blue Calculate button to kick off the calculator. The results show up in the right panel.

Step 7. If you wish to compare a data set, change the input parameters, and press the Calculate button. In this example, a muzzle device was added.

Input Parameters

Unique Name:

308 WIN BRAKE

Cartridge:

308 Win

Firearm Class:

rifle: standard

Action Type:

bolt action

Gun Weight:

8.0

lbs

Bullet Weight:

125

grains

Muzzle Velocity:

2973

fps

Powder Charge:

46.5

grains

Muzzle Device:

brake: moderate

Stock Fit

LOP:

just right

Comb Height:

perfect cheek weld

Recoil Pad:

soft recoil pad

Calculate

Save Data Set

Results

Data Set 1

Data Set 2

Data Set 3

Unique Name:

308 WIN

308 WIN BRAKE

--

Cartridge Selected:

308 Win

308 Win

--

Firearm Class:

rifle: standard

rifle: standard

--

Action Type:

bolt action

bolt action

--

Gun Weight:

8.0 lbs

8.0 lbs

--

Bullet Weight:

125.0 gr

125.0 gr

--

Muzzle Velocity:

2973 fps

2973 fps

--

Powder Charge:

46.5 gr

46.5 gr

--

Muzzle Device:

none

brake: moderate

--

LOP:

just right

just right

--

Comb Height:

perfect cheek weld

perfect cheek weld

--

Recoil Pad:

soft recoil pad

soft recoil pad

--

Calculated Results

Free Recoil Energy

14.92 ft-lbs

10.45 ft-lbs

-- ft-lbs

Recoil Velocity

10.96 fps

7.67 fps

-- fps

Avg Impulse Force

1434 lbf

1004 lbf

-- lbf

Perceived Recoil Score

5.7 /10

4.4 /10

-- /10





Step 8. Press the green Save Data Set button if you wish to refer back to a specific firearm – ammunition combination without having to reenter the data.

The screenshot shows the Bcalc Recoil Calculator application window. The 'Input Parameters' section on the left contains the following fields:

- Unique Name: 308 WIN BRAKE SA
- Cartridge: 308 Win
- Firearm Class: rifle: standard
- Action Type: semi auto
- Gun Weight: 8.0 lbs
- Bullet Weight: 125 grains
- Muzzle Velocity: 2973 fps
- Powder Charge: 46.5 grains
- Muzzle Device: brake: moderate
- Stock Fit: LOP: just right, Comb Height: perfect cheek weld, Recoil Pad: soft recoil pad

The 'Results' section on the right displays three data sets:

	Data Set 1	Data Set 2	Data Set 3
Unique Name:	308 WIN	308 WIN BRAKE	308 WIN BRAKE SA
Cartridge Selected:	308 Win	308 Win	308 Win
Firearm Class:	rifle: standard	rifle: standard	rifle: standard
Action Type:	bolt action	bolt action	semi auto
Gun Weight:	8.0 lbs	8.0 lbs	8.0 lbs
Bullet Weight:	125.0 gr	125.0 gr	125.0 gr
Muzzle Velocity:	2973 fps	2973 fps	2973 fps
Powder Charge:	46.5 gr	46.5 gr	46.5 gr
Muzzle Device:	none	brake: moderate	brake: moderate
Stock Fit:	just right	just right	just right
Comb Height:	perfect cheek weld	perfect cheek weld	perfect cheek weld
Recoil Pad:	soft recoil pad	soft recoil pad	soft recoil pad

The 'Calculated Results' section at the bottom right shows the following values:

	Data Set 1	Data Set 2	Data Set 3
Free Recoil Energy	14.92 ft-lbs	10.45 ft-lbs	10.45 ft-lbs
Recoil Velocity	10.96 fps	7.67 fps	7.67 fps
Avg Impulse Force	1434 lbf	1004 lbf	829 lbf
Perceived Recoil Score	5.7 /10	4.4 /10	3.9 /10

A 'Saved' dialog box is open in the center, displaying the message 'Saved successfully (ID: 2).'. The 'Calculate' button is blue, and the 'Save Data Set' button is green. A small illustration of a chicken is visible at the bottom left of the results section.

5. Credits & License

The Bcalc Recoil Calculator was created by Brian Calc.

GNU General Public License (GPL) Version 3.